AETHER LMS

30-DAY ENGINEERING DEVELOPMENT ROADMAP

Coursera Usability x NPTEL Academic Rigor -- Executive Architecture Report

AUTHOR	TIMELINE	TECH STACK	STATUS
Principal System Architect	**30 Days (4 Sprints)**	**Next.js 14, Supabase, Postgres**	**READY FOR DEV**

1. EXECUTIVE SUMMARY & PLATFORM OBJECTIVES

AetherLMS bridges the critical divide in modern higher education by unifying Coursera's seamless video streaming and modular navigation with NPTEL's rigid deadlines, automated assignment scoring, best-of-N algorithms (e.g. top 8 of 12 weeks), proctored exams, and accredited QR verification.

COURSERA LIMITS
- Soft deadlines (12% completion)
- No multi-week drop grading
- No proctored exam scheduling

NPTEL LIMITS
- Legacy, clunky video player
- No synced transcript seek
- Slow, manual TA grading loop

AETHERLMS ADVANTAGE
+ 0.5x-2x player with seek sync
+ Best-8-of-12 automatic engine
+ Tamper-proof QR certificates

2. 30-DAY MASTER SPRINT CALENDAR

PHASE	FOCUS	KEY DELIVERABLES	SECURITY & RLS
Week 1 (D1-7)	**DB, Auth & Role Security**	16-table Postgres schema, SSR Auth, Edge RBAC middleware	**4 Roles Protected**
Week 2 (D8-14)	**Next-Gen Content Engine**	HLS Player, Synced Transcript Seek, Drag-and-drop Studio	**Enrolled Isolation**
Week 3 (D15-21)	**Evaluation & Best-of-N**	Anti-cheat Quiz, Hard cutoff Labs, Best-of-8 trigger, TA Queue	**Hard Cutoff Limits**
Week 4 (D22-30)	**Verification, QA & Launch**	PDF Certificates with QR, Playwright E2E, Vercel Production	**Public Validation**

CORE ARCHITECTURAL INNOVATIONS

* Dynamic Transcript-to-Video Seek: Video scrubber binds bi-directionally to sentence-level cues.
* Deterministic Drop-Grading Engine: PostgreSQL triggers automatically reconcile grades upon TA review.
* Tamper-Proof Credential Hash: Every issued certificate embeds SHA-256 validation for employers.

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3. DETAILED DAY-BY-DAY EXECUTION MATRIX

WEEK 1: FOUNDATION, DATABASE ARCHITECTURE & ROLE SECURITY (DAYS 1-7)

Days 1-2: 16-Table PostgreSQL Schema & Row Level Security (RLS)

- Deploy profiles, courses, modules, lessons, submissions, quizzes, certificates, and exam tables.
- Strict RLS isolation: students access enrolled material only; instructors manage own courses.

Days 3-4: Next.js 14 App Router & Global Layout Architecture

- Tailwind custom palette, Shadcn UI primitives, TanStack Query provider, and React Hot Toast.
- Responsive layout shells for Student Dashboard, Instructor Studio, and Admin Control Panel.

Days 5-7: Authentication, Session Management & RBAC Middleware

- Email/Password and OAuth (Google/GitHub) PKCE flows via @supabase/ssr.
- Edge middleware route guards for /dashboard, /instructor, /admin, and /course/learn.

WEEK 2: NEXT-GEN CONTENT ENGINE -- BEYOND COURSERA (DAYS 8-14)

Days 8-10: Interactive Video Player & Synced Transcript Seek

- HLS streaming player with 0.5x to 2x speed toggles, auto-resume position, and auto-advance.
- Side-by-side interactive transcript with click-to-seek jumping directly to the video second.

Days 11-12: Curriculum Builder Studio & Supabase Storage

- Drag-and-drop course builder across Modules, Weeks, and Lessons with prerequisite locks.

Days 13-14: Course Discovery Catalog & One-Click Enrollment

- Faceted search (Category, Difficulty, Rating) and 12-week syllabus overview accordion.

WEEK 3: RIGOROUS EVALUATION & ASSIGNMENT ENGINE -- NPTEL EVOLUTION (DAYS 15-21)

Days 15-17: Anti-Cheat Timed Quiz & Rigid Deadline Lab Submissions

- Full-screen countdown quiz with Question Palette (Answered/Flagged) and auto-submit on timeout.
- Rigid deadline countdown clocks with hard cutoffs, inline Go/C code editor, and SHA-256 receipts.

Days 18-19: Best-of-N Flexible Score Engine (PostgreSQL Trigger)

- Automated trigger calculate_best_of_n_score() drops lowest 4 weeks, averaging top 8 of 12.

Days 20-21: TA Gradebook Queue & Real-Time Q&A Forum

- Evaluator queue with CI test results, rubric sliders, and Realtime verified instructor badges.

WEEK 4: VERIFICATION, PERFORMANCE, TESTING & LAUNCH (DAYS 22-30)

Days 22-23: Dynamic PDF Certificates & Public QR Verification

- Automated PDF certificate generation (@react-pdf/renderer) for scores >= 75%.
- Public route /verify/[id] validating recipient, grade breakdown, and security hash via QR.

Days 24-25: Query Performance Optimization & Mobile Viewport Audit

- B-tree indexing on foreign keys and touch-friendly mobile classroom interface.

Days 26-30: Autonomous E2E Tests, Vercel Production & Go-Live

- Playwright E2E test suites covering Student, Instructor, and Admin user journeys.
- Production deployment on Vercel with custom domain, Supavisor pooling, and Sentry monitoring.